

Alumni Management System — User Interface Design

150 Minutes • HTML5 + CSS3 • Project-Based Practical

Lab Duration	Technology	Project
150 Minutes	HTML5 + CSS3	Alumni Management System

1. Lab Overview

In this practical lab, students will design the front-end user interface of an Alumni Management System using HTML5 and CSS3. The work is component-based: students build a navigation bar, dashboard, statistics cards, alumni profile, directory table and registration form.

2. Learning Objectives

- Create a professional web page layout using CSS.
- Apply selectors, typography, colors, spacing, borders, shadows, and the CSS box model.
- Use Flexbox to build navigation and component layouts.
- Use CSS Grid to create responsive dashboard cards.
- Design reusable cards, buttons, forms, tables, and profile components.
- Use pseudo-classes such as :hover and :focus.
- Organize CSS into a reusable external stylesheet.

3. Duration Plan

Activity	Time
Introduction & requirements	10 min
Exercise 1 — Project setup	10 min
Exercise 2 — Main page layout	20 min
Exercise 3 — Navigation bar	15 min
Exercise 4 — Dashboard cards	20 min
Exercise 5 — Alumni profile card	20 min
Exercise 6 — Alumni directory	20 min
Exercise 7 — Registration form	30 min

4. Project Structure

```
alumni-management/  
├── index.html  
├── alumni.html  
├── profile.html  
├── register.html  
└── css/  
    └── style.css
```

Rule For this lab, do not use Bootstrap or Tailwind CSS. The objective is to practice writing CSS yourself. Use an external stylesheet rather than inline CSS.

Exercise 1 — Project Setup

Time: 10 Minutes

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>Alumni Management System</title>  
  <link rel="stylesheet" href="css/style.css">  
</head>  
<body>  
  <header>Alumni Management System</header>  
  <main>  
    <h1>Welcome to Alumni Management System</h1>  
  </main>  
</body>  
</html>
```

Student Tasks

- Set a global font.
- Remove the default body margin.
- Add a page background color.
- Style the header.
- Center the main heading.

Key Concepts CSS selectors, font-family, background, margin, padding, text-align, color.

Exercise 2 — Create the Main Page Layout

Time: 20 Minutes

Student Tasks

- Create a header, navigation, main content area, dashboard section, statistics section, and recent alumni section.
- Use semantic HTML: <header>, <nav>, <main>, <section>, and <footer>.
- Apply width, max-width, margin, padding, background, border, and box-shadow.

Key Concepts Page structure and the CSS box model.

Exercise 3 — Design the Navigation Bar

Time: 15 Minutes

```
<nav class="navbar">
  <div class="logo">AlumniConnect</div>
  <ul class="nav-links">
    <li><a href="#">Dashboard</a></li>
    <li><a href="#">Alumni</a></li>
    <li><a href="#">Events</a></li>
    <li><a href="#">Profile</a></li>
    <li><a href="#">Logout</a></li>
  </ul>
</nav>
```

Student Tasks

- Use Flexbox to place the logo left and menu items right.
- Align items vertically and add spacing.
- Remove list bullets.
- Add hover effects to links.

Key Concepts display:flex, justify-content, align-items, gap, :hover.

Exercise 4 — Dashboard Statistics Cards

Time: 20 Minutes

```
<section class="stats">
  <div class="stat-card"><h3>Total Alumni</h3><p>1,250</p></div>
  <div class="stat-card"><h3>Events</h3><p>25</p></div>
  <div class="stat-card"><h3>Jobs Posted</h3><p>87</p></div>
</section>
```

Student Tasks

- Use CSS Grid.
- Desktop: four cards in one row.
- Tablet: two cards per row.
- Mobile: one card per row.

- Style cards with spacing, border radius, and shadow.

Key Concepts display:grid, grid-template-columns, gap, responsive layout.

Exercise 5 — Alumni Profile Card

Time: 20 Minutes

```
<div class="profile-card">
  <div class="profile-image">
    
  </div>
  <h2>Muhammad Ahmed</h2>
  <p class="designation">Software Engineer</p>
  <div class="profile-info">
    <p><strong>Batch:</strong> 2022</p>
    <p><strong>Department:</strong> Computer Science</p>
    <p><strong>City:</strong> Karachi</p>
  </div>
  <div class="profile-actions">
    <button>View Profile</button>
    <button>Contact</button>
  </div>
</div>
```

Student Tasks

- Style the profile image.
- Use padding, border-radius, and box-shadow.
- Style the buttons.
- Add a hover effect to the card.
- Keep the component visually balanced and consistent.

Key Concepts Cards, images, buttons, hover effects, spacing.

Exercise 6 — Alumni Directory

Time: 20 Minutes

```
<section class="directory">
  <h1>Alumni Directory</h1>
  <div class="search-box">
    <input type="text" placeholder="Search alumni...">
    <button>Search</button>
  </div>
  <table>
    <thead>
      <tr><th>Name</th><th>Batch</th><th>Department</th></tr>
    </thead>
    <tbody>
      <tr><td>Ali Khan</td><td>2022</td><td>Computer Science</td></tr>
      <tr><td>Ahmed Raza</td><td>2021</td><td>Software Engineering</td></tr>
```

```

        <tr><td>Sara Ahmed</td><td>2023</td><td>Information Technology</td></tr>
    </tbody>
</table>
</section>

```

Student Tasks

- Style the search input and button.
- Style the table header and rows.
- Add borders and readable spacing.
- Highlight an alumni row on hover.

Key Concepts Forms, tables, spacing, tr: hover.

Exercise 7 — Alumni Registration Form

Time: 15 Minutes

```

<section class="form-container">
  <h1>Alumni Registration</h1>
  <label>Full Name</label>
  <input type="text">

  <label>Email</label>
  <input type="email">

  <label>Batch</label>
  <input type="number">

  <label>Department</label>
  <select>
    <option>Computer Science</option>
    <option>Software Engineering</option>
    <option>Information Technology</option>
  </select>

  <label>Current Organization</label>
  <input type="text">

  <button>Register</button>
</section>

```

Student Tasks

- Style the form container and labels.
- Style input, select, and button elements.
- Create a clear focus state using input: focus.
- Make the submit button visually prominent.

Key Concepts Form styling, pseudo-classes, focus states.

5. Target Interface

AlumniConnect

Profile / Logout

Dashboard | Alumni | Events | Jobs | Profile

Welcome Back!

Manage your alumni network

[Total Alumni] [Events] [Jobs]

Name	Batch	Department
Ali Khan	2022	Computer Science
Sara Ahmed	2023	IT

6. CSS Topics Covered

CSS Topic	Application
CSS Selectors	Styling UI elements
Colors & Typography	Application theme and readable text
Box Model	Cards and containers
Margin & Padding	Component spacing
Borders & Border Radius	Modern UI components
Box Shadow	Card depth
Flexbox	Navigation and alignment
CSS Grid	Dashboard layout
Pseudo-classes	Hover and focus states
Forms	Alumni registration
Tables	Alumni directory

8. Submission Requirements

- Submit index.html, alumni.html, profile.html, register.html, and css/style.css.
- All pages must be linked together.
- CSS must be stored in an external stylesheet; no inline CSS.
- Use meaningful class names and consistent formatting.
- Use Flexbox for navigation and CSS Grid for dashboard cards.
- Include hover and focus effects.
- The final interface should be professional, readable, and responsive.

9. Learning Outcome

After completing this 150-minute practical, students will have built the front-end UI of a small Alumni Management System using HTML5 and CSS3. They will demonstrate practical competence in page layout, Flexbox, CSS Grid, forms, tables, cards, pseudo-classes and reusable UI components.